

Submission PY1-MID-SUB08

Question PY1-MID-Q01

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visible work:

range(4) gives 0 through 3... ?

last line is hard to read; maybe [0, 4]

Question PY1-MID-Q02

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visible work:

counts = Counter(xs)... ?

last line is hard to read; maybe scan counts, then scan in order; return first count==1 else None; $O(n)$ time, $O(n)$ space

Question PY1-MID-Q03

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visible work:

A single pass fixes only adjacent inversions encountered once.... ?

last line is hard to read; maybe repeat passes or track the shrinking unsorted suffix; one pass alone is not a full sort

Question PY1-MID-Q04

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visible work:

linear scan: up to 64 comparisons, $\Theta(n)$... ?

last line is hard to read; maybe linear $\Theta(n)$, up to 64; binary $\Theta(\log n)$, about 7

Question PY1-MID-Q05

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visible work:

Removing shifts later elements left while the iterator advances.... ?

last line is hard to read; maybe mutation shifts elements and skips checks; use a filtered copy or iterate over a copy

Question PY1-MID-Q06

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visible work:

range(4) gives 0 through 3... ?

last line is hard to read; maybe [0, 4]

Question PY1-MID-Q07

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visible work:

counts = Counter(xs)... ?

last line is hard to read; maybe scan counts, then scan in order; return first count==1 else None; $O(n)$ time, $O(n)$ space

Question PY1-MID-Q08

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visible work:

A single pass fixes only adjacent inversions encountered once.... ?

last line is hard to read; maybe repeat passes or track the shrinking unsorted suffix; one pass alone is not a full sort